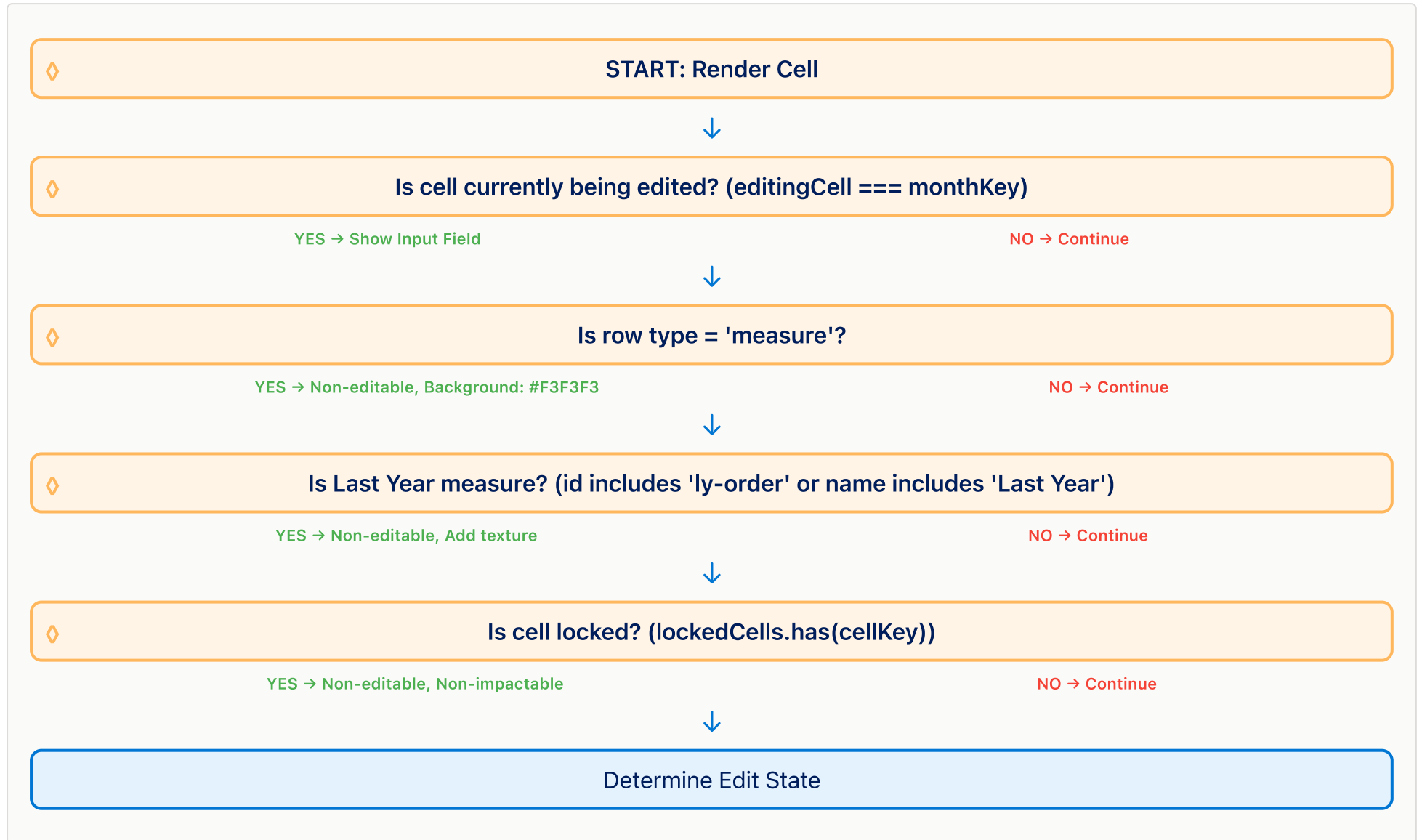


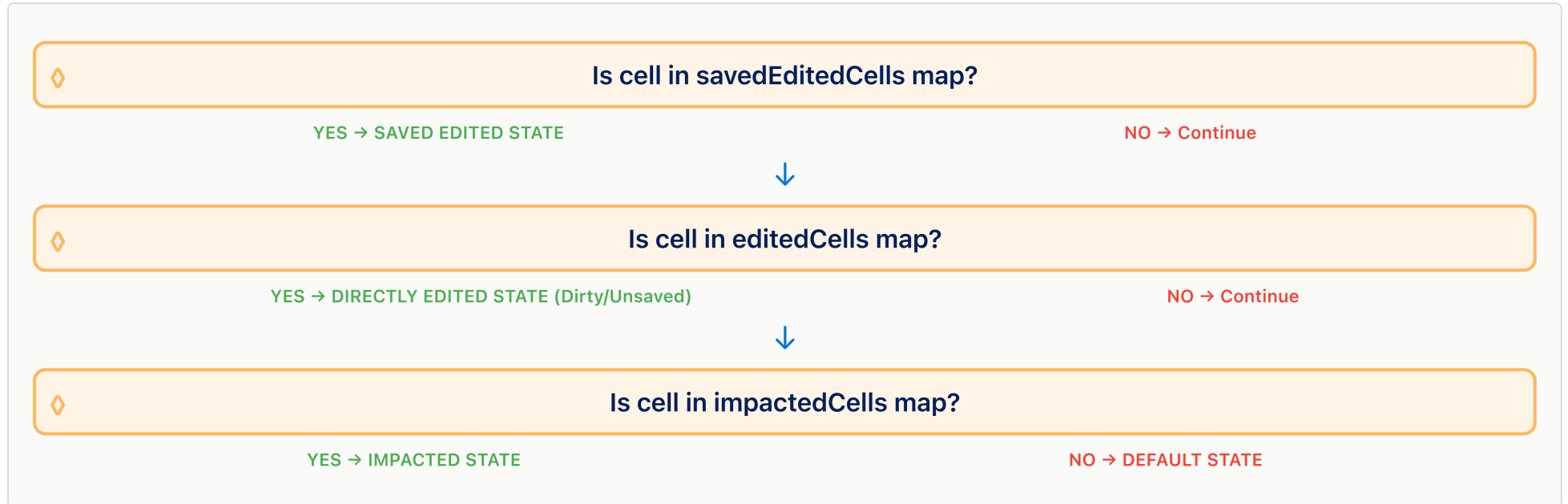
Cell State Decision Framework

Complete Flowchart & Condition Reference for Forecasting & Planning Grid

1. Primary State Determination Flow



2. Edit State Determination Flow



Important: States are mutually exclusive in this order: Saved Edited > Directly Edited > Impacted > Default

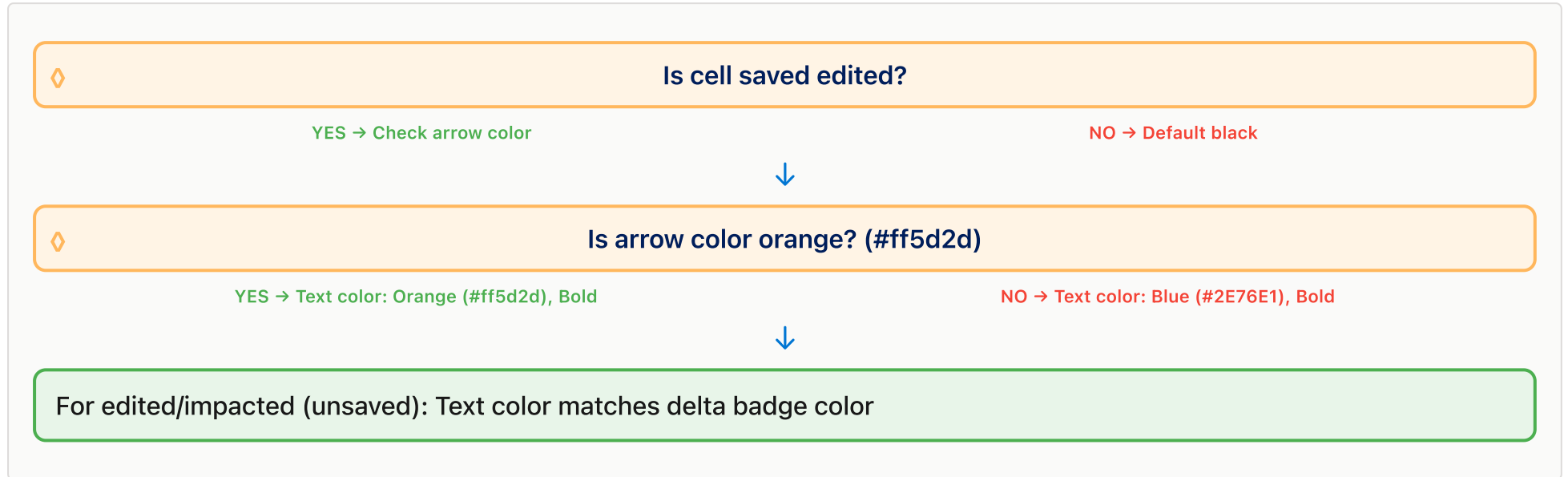
3. Visual Indicator Priority Rules

Priority	Indicator	Condition	Replaces	Visual
1	Lock Icon	<code>isCellLocked === true</code>	Arrow icon	Grey lock ( #6b7280)
2	Arrow Icon	<code>isSavedEdited === true && !isCellLocked</code>	Delta badge	↑ Orange (#ff5d2d) or ↓ Blue (#2E76E1)
3	Delta Badge	<code>(isDirectlyEdited isImpacted) && !isSavedEdited</code>	None	Percentage change badge
4	Note Triangle	<code>hasNote === true</code>	None (overlay)	Top-right corner ( #DA9401)
5	Search Highlight	<code>valueMatchesSearch === true</code>	None (text highlight)	Yellow background on text ( #FFE066)

4. Background Color Decision Matrix

Condition	Background Color	Color Code	Exception
Row type = 'measure'	☐ Light Grey	#F3F3F3	Always applies, even if edited/impacted
Row type = 'time' (in Time/Dimensions layout)	☐ Light Grey	#F3F3F3	Always applies
<code>isDirectlyEdited === true</code>	☒ Light Orange	#f9e3b6	Unless in measure row
<code>isImpacted === true</code>	☒ Light Yellow	#fef5e7	Unless in measure row (keeps #F3F3F3)
<code>isSavedEdited === true</code>	☐ White	#ffffff	Unless in measure row
Default state	☐ White	#ffffff	Standard unedited cell

5. Text Color Decision Flow



6. Complete State Combination Matrix

Edit State	Locked	Has Note	Row Type	Background	Icon	Text Color	Editable
Default	No	No	Dimension	White	None	Black	✓
Default	No	Yes	Dimension	White	None	Black	✓
Default	No	No	Measure	#F3F3F3	None	Black	×
Directly Edited	No	No	Dimension	#f9e3b6	None	Orange/Blue	✓
Directly Edited	No	Yes	Dimension	#f9e3b6	None + Triangle	Orange/Blue	✓
Directly Edited	Yes	No	Dimension	#f9e3b6	Lock	Orange/Blue	×
Saved Edited	No	No	Dimension	White	Arrow	Orange/Blue Bold	✓
Saved Edited	No	Yes	Dimension	White	Arrow + Triangle	Orange/Blue Bold	✓
Saved Edited	Yes	No	Dimension	White	Lock	Orange/Blue Bold	×
Impacted	No	No	Dimension	#fef5e7	None	Orange/Blue	✓
Impacted	No	Yes	Dimension	#fef5e7	None + Triangle	Orange/Blue	✓
Impacted	No	No	Measure	#F3F3F3	None	Orange/Blue	×
Impacted	Yes	No	Dimension	#fef5e7	Lock	Orange/Blue	×
Default	No	No	Readonly	White + Texture	None	Black	×

7. Edge Cases & Special Conditions

Condition	Behavior
Cell in editedCells AND impactedCells	Impossible - impactedCells only populated if NOT in editedCells. Priority: editedCells > impactedCells
Locked cell in measure row	Shows lock icon, retains measure row background (#F3F3F3), non-editable, non-impactable
Saved edited cell gets impacted	Remains in savedEditedCells, shows arrow, but value updates. Arrow color based on original edit direction.
Cell with note but no value change	Shows note triangle, white background, no arrow/badge, editable
Search matches edited cell	Shows yellow highlight on matching text, retains edited cell styling (orange background, delta badge)
Focused edited cell	Shows black outline (2px), retains edited styling, can show edit info popover
Dirty cell (edited but not saved)	No arrow shown, only delta badge. Arrow appears after save.
Delta = 0%	Delta badge not shown if deltaPercent < 0.001
Locked cell contributes to sums	Yes - locked cells contribute current value to parent sums and aggregations, but don't receive propagation changes
Impacted cell in measure row	Retains measure row background (#F3F3F3) instead of yellow (#fef5e7)

8. State Transition Rules

From State	Action	To State	Visual Change
Default	User edits value	Directly Edited	Background → Orange, Delta badge appears
Directly Edited	User saves grid	Saved Edited	Background → White, Badge → Arrow, Text → Colored & Bold
Default	Another cell edited (propagation)	Impacted	Background → Yellow, Delta badge appears
Any	User locks cell	Locked	Icon → Lock (replaces arrow), Non-editable
Any	User adds note	Same + Note	Triangle appears (no other change)
Directly Edited	User cancels	Default	Background → White, Badge removed
Impacted	User edits impacted cell	Directly Edited	Background → Orange (from Yellow)

9. Missing Conditions Checklist

Verified Conditions:

- ✓ Currently editing state (input field shown)
- ✓ Row type determination (measure/dimension/time)
- ✓ Readonly measure detection (Last Year)
- ✓ Lock state detection
- ✓ Edit state priority (saved > edited > impacted > default)
- ✓ Note detection (from editHistory)
- ✓ Search highlighting
- ✓ Focus state
- ✓ Hover state (row level, not cell level)
- ✓ Delta calculation (percentage)
- ✓ Arrow direction (increase/decrease)
- ✓ Measure row background override
- ✓ Locked cell propagation protection

Forecasting & Planning Tool - Cell State Decision Framework

Generated: December 29, 2025